

MountainCar Q-Learning - Quick Start Guide

What You'll Do

Train an AI agent to drive a car up a steep hill by building momentum!

Option 1: Automated Setup (Easiest - Mac/Linux)

```
bash

# 1. Download all files to a folder called 'mountaincar_qlearning'
# 2. Open terminal in that folder
# 3. Run this one command:

chmod +x setup_and_run.sh && ./setup_and_run.sh
```

That's it! Script does everything automatically.

Option 2: Manual Setup (Works on All Systems)

Step 1: Create Project Directory

```
bash

mkdir mountaincar_qlearning
cd mountaincar_qlearning
```

Step 2: Create Virtual Environment

```
bash

python3 -m venv venv
```

Step 3: Activate Virtual Environment

Mac/Linux:

```
bash

source venv/bin/activate
```

Windows:

```
cmd

venv\Scripts\activate
```

You should see `(venv)` in your terminal prompt.

Step 4: Install Required Packages

```
bash

pip install gymnasium numpy matplotlib
```

Step 5: Add the Code File

Save `mountaincar_qlearning.py` in the `mountaincar_qlearning` folder.

Step 6: Run the Training

```
bash

python mountaincar_qlearning.py
```

What to Expect

Training Output:

```
=====
TRAINING MOUNTAINCAR Q-LEARNING AGENT
=====

Episode  100/10000 | Avg Reward: -200.00 | Avg Steps: 200.00 | Epsilon: 0.951
Episode  200/10000 | Avg Reward: -200.00 | Avg Steps: 200.00 | Epsilon: 0.905
Episode  300/10000 | Avg Reward: -200.00 | Avg Steps: 200.00 | Epsilon: 0.861
...
[Around episode 3000-5000, you'll see improvement:]
...
Episode  5000/10000 | Avg Reward: -165.32 | Avg Steps: 165.32 | Epsilon: 0.082
Episode  6000/10000 | Avg Reward: -142.45 | Avg Steps: 142.45 | Epsilon: 0.050
Episode  7000/10000 | Avg Reward: -118.67 | Avg Steps: 118.67 | Epsilon: 0.030
Episode  8000/10000 | Avg Reward: -105.23 | Avg Steps: 105.23 | Epsilon: 0.018
Episode  9000/10000 | Avg Reward: -98.45 | Avg Steps: 98.45 | Epsilon: 0.011
Episode 10000/10000 | Avg Reward: -96.78 | Avg Steps: 96.78 | Epsilon: 0.010
=====
TRAINING COMPLETED
=====
```

Generated Files:

- `mountaincar_training_results.png` - 6 plots showing training progress

Test Results:

```
Testing trained agent...
Test Episode  1: Reward = -98.00, Steps = 98
Test Episode  2: Reward = -95.00, Steps = 95
Test Episode  3: Reward = -102.00, Steps = 102
```

...

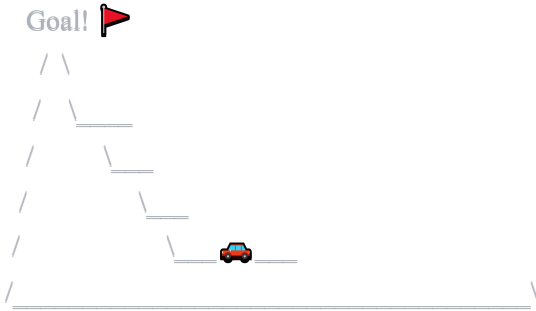
Average Test Reward: -98.50

Average Test Steps: 98.50

Agent successfully learned to reach the goal efficiently!

Understanding MountainCar

The Problem:



- Car starts in a valley
- Goal: Reach the flag on the right hill
- Problem: Engine too weak to drive straight up!
- Solution: Rock back and forth to build momentum

Key Differences from CartPole:

Feature	CartPole	MountainCar
Goal	Balance pole	Reach flag
Reward	+1 per step (good!)	-1 per step (sparse!)
Actions	2 (left, right)	3 (left, none, right)
Success	195+ steps average	<110 steps average
Difficulty	Medium	Hard
Training time	~3k episodes	~10k episodes

Why It's Harder:

1. **Sparse rewards:** Every action gives -1, no positive feedback
2. **Counter-intuitive:** Must go LEFT (away from goal) to build momentum
3. **Strategic:** Requires planning multiple steps ahead
4. **Delayed reward:** Good actions now pay off 10+ steps later

Interpreting Results

Reward Values:

- `(-200)`: Bad (timeout, didn't reach goal)
- `(-150)`: Getting better (reached goal slowly)
- `(-110)`: Good! (reached goal efficiently)
- `(-95)`: Excellent! (near optimal)

Remember: Less negative = better!

Success Criteria:

- ✓ Average reward (last 100 episodes): -95 to -110
 - ✓ Test performance: < 110 steps consistently
 - ✓ Plots show clear improvement trend
-

Troubleshooting

"Always getting -200 rewards"

- **Normal for first 3000-5000 episodes!**
- Agent needs time to stumble upon success
- Keep training, breakthrough will come

"Training is slow"

- Expected: ~5-10 minutes for 10,000 episodes
- If >20 minutes, check if computer is doing other tasks

"ImportError: No module named gymnasium"

- Make sure venv is activated (see `(venv)` in prompt)
- Run: `pip install gymnasium numpy matplotlib`

"Results are inconsistent"

- Normal! MountainCar is harder than CartPole
 - Some randomness in exploration
 - Average over 100 episodes is what matters
-

What the Plots Show

Plot 1: Reward per Episode

- Raw rewards for each episode
- Should trend upward (less negative)

- Will be noisy (lots of variance)

Plot 2: Moving Average Reward

- Smoothed trend
- Should clearly show improvement
- Target: reach -110 or better

Plot 3: Episode Length (Steps)

- How many steps to reach goal
- Should trend downward
- Target: under 110 steps

Plot 4: Moving Average Steps

- Smoothed steps trend
- Clearer view of learning

Plot 5: Epsilon Decay

- Exploration rate over time
- Starts at 1.0, decays to 0.01

Plot 6: Reward Distribution

- Histogram of all episode rewards
- Should shift right (less negative) over training

Files You Have

1. **mountaincar_qlearning.py** (5.2 KB)
 - Clean code, no comments
 - Ready to run
 2. **MOUNTAINCAR_QLEARNING_GUIDE.txt** (22 KB)
 - Complete conceptual explanation
 - Comparison with CartPole
 - Troubleshooting guide
 3. **setup_and_run.sh** (Optional)
 - Automated setup script
 - Mac/Linux only
-

Quick Commands Reference

```
bash

# Setup (one time)
python3 -m venv venv
source venv/bin/activate # or venv\Scripts\activate on Windows
pip install gymnasium numpy matplotlib

# Run training
python mountaincar_qlearning.py

# Deactivate venv when done
deactivate
```

Time Estimates

- Setup: 2 minutes
- Training: 5-10 minutes
- Reviewing results: 5 minutes
- **Total: ~15-20 minutes**

Success Checklist

After running, you should have:

- ☒ Training completed (10,000 episodes)
- ☒ Plot file: `mountaincar_training_results.png`
- ☒ Final average reward: -95 to -110
- ☒ Test performance: < 110 steps average
- ☒ Clear upward trend in plots

If all checked, you've successfully trained a MountainCar agent! 🎉

Next Steps (Optional)

1. **Experiment:** Try different hyperparameters
 2. **Compare:** How does it differ from CartPole?
 3. **Analyze:** Why did certain strategies work?
 4. **Visualize:** Watch the agent in action (add visualization code)
-

Need Help?

Read the comprehensive guide: [MOUNTAINCAR_QLEARNING_GUIDE.txt](#)

It covers:

- Detailed problem explanation
- Why MountainCar is hard
- Expected training phases
- Troubleshooting common issues
- Comparison with CartPole

Good luck! Remember: seeing -200 for thousands of episodes is NORMAL. Be patient! 🚗 ⚙️